A MICRO PROJECT REPORT ON

MOGO – THE HOME AUTOMATION CONTROLLER

Submitted for partial fulfillment of the requirements for the subject PBCST504 MICROCONTROLLERS

FOR

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

(CYBER SECURITY)

SUBMITTED BY

JOYAL JOSHY – SNM24CC005

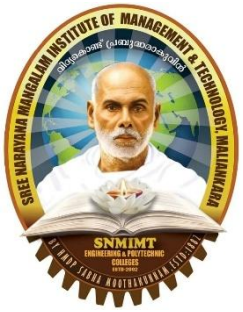
KAVYA R KRISHNA – SNM24CC006

MOHAMMED YASIN P SAJIL – SNM24CC007

RUTH ALOSHIOUS – SNM24CC008

Under the supervision of

Ms. ASWATHI C D



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

SREE NARAYANA MANGALAM INSTITUTE OF MANAGEMENT & TECHNOLOGY, MALIANKARA

AFFILIATED TO APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY, KERALA

(2026–2027)

SREE NARAYANA MANGALAM INSTITUTE OF MANAGEMENT & TECHNOLOGY, MALIANKARA

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

CERTIFICATE

This is to certify that the report entitled "MOGO – The Home Automation Controller: A Detachable Gesture-Controlled Home Automation Robo", submitted by JOYAL JOSHY (SNM24CC005), KAVYA R KRISHNA (SNM24CC006), MOHAMMED YASIN P SAJIL (SNM24CC007) and RUTH ALOSHIOUS (SNM24CC008) to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY) is a bonafide record of the project work carried out by them under my guidance and supervision.

SUPERVISING GUIDE
Ms. ASWATHI C D

DECLARATION

We, the undersigned, hereby declare that the project report "MOGO – The Home Automation Controller: A Detachable Gesture-Controlled Home Automation Robo" submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology in Computer Science and Engineering (Cyber Security) of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under the supervision of Ms. ASWATHI C D. This submission represents our ideas in our own words and where ideas or words of others have been included; we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

NAME OF THE STUDENT:

JOYAL JOSHY – SNM24CC005
KAVYA R KRISHNA – SNM24CC006
MOHAMMED YASIN P SAJIL – SNM24CC007
RUTH ALOSHIOUS – SNM24CC008

DATED SIGNATURE:

Place: MALIANKARA
Date: 08/10/2026

ACKNOWLEDGEMENT

We express our heartfelt gratitude to the almighty, the supreme guide for bestowing his blessings in our entire endeavor.

We wish to give deep sense of acknowledgement to our Principal Dr. ATMARAM.P, for his constant encouragement and valuable advice throughout the course.

We are deeply indebted to Mrs. Lakshmi, Head of the Department of COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY), for her sincere and dedicated cooperation and encouragement throughout the duration of our project.

We would also like to thank our project guide Ms. Aswathi C D, Department of COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY) for her valuable advice and whole hearted cooperation without which this project would not have seen the light of the day. We express our sincere gratitude to all the faculty members of the Department of COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY) for their co-operation and support to our project.

We are highly obliged to our parents for their encouragement and we express our sincere thanks to each and every one who has directly or indirectly helped us during the period of this project.

ABSTRACT

MOGO – The Home Automation Controller is a portable, handheld smart-home interface designed to provide natural, seamless, and convenient control of connected household appliances. Modern home automation systems frequently rely on conventional physical switches, handheld infrared remotes, or smartphone-based applications. While functional, these traditional approaches introduce operational constraints such as requiring direct line-of-sight, navigating through multi-step mobile operating systems, or locating specific physical remotes in low-light environments.

To overcome these limitations, MOGO proposes an integrated, multimodal embedded architecture that unifies motion-based gesture input, capacitive touch input, and digital audio communication into an ergonomic handheld controller. An ESP32-C3 Super Mini acts as the central controller and communication unit, providing Wi-Fi connectivity and managing an event-driven FreeRTOS firmware pipeline. An MPU6050 six-axis accelerometer and gyroscope module senses hand movement and orientation, allowing predefined kinematic gestures to be translated into appliance commands. A TTP223 capacitive touch sensor provides an instantaneous direct input trigger, while an INMP441 omnidirectional digital MEMS microphone supports voice input via I2S.

Immediate system feedback is provided through visual and auditory channels: an SSD1306 OLED screen displays operational modes and connectivity status over I2C, while a MAX98357A I2S Class-D amplifier drives a 4 Ohm, 3 W speaker for acoustic prompts and confirmations. The controller operates autonomously from a rechargeable 3.75 V Li-Po battery charged through a TP4056 USB charging management module. Comprehensive functional testing demonstrates that the prototype successfully integrates gesture, touch, voice, Wi-Fi communication, and multimedia feedback into a single, compact embedded home-automation platform.

CONTEXT

Chapter 1: INTRODUCTION 1
1.1 Project Overview
1.2 Problem Statement
1.3 Objectives
1.4 Scope of the Project
Chapter 2: LITERATURE REVIEW 3
2.1 Conventional Home Automation Controllers
2.2 Gesture-Based Interaction
2.3 IoT and Wireless Home Automation
2.4 Multimodal User Interfaces
Chapter 3: SYSTEM REQUIREMENTS 5
3.1 Hardware Requirements (ESP32-C3, MPU6050, OLED, I2S)
3.2 Software Requirements (ESP-IDF, FreeRTOS, Arduino)
Chapter 4: SYSTEM DESIGN AND ARCHITECTURE 7
4.1 System Architecture
4.2 Working Principle
4.3 System Flow
Chapter 5: IMPLEMENTATION 9
5.1 Hardware Integration
5.2 Gesture and Touch Processing
5.3 Voice and Audio Interface
5.4 Wireless Communication
Chapter 6: TESTING 11
6.1 Testing Methodology
6.2 Test Cases
6.3 Performance Parameters
Chapter 7: RESULTS AND DISCUSSION 13
7.1 Functional Testing Results
7.2 Results
7.3 Analysis
7.4 Key Findings
Chapter 8: CONCLUSION AND FUTURE SCOPE 15
8.1 Conclusion
8.2 Future Scope
REFERENCES 16
APPENDICES
Appendix A: Component Summary 17
Appendix B: System Flow & Architecture 18

Chapter 1: INTRODUCTION

1.1 Project Overview

Home automation systems are increasingly deployed to enhance living comfort, energy management, and physical accessibility across modern residential and commercial environments. Traditional appliance control has predominantly depended on conventional wall switches, handheld infrared (IR) remotes, or smartphone applications. While infrared remotes require uninterrupted line-of-sight and physical button manipulation, smartphone applications necessitate unlocking the mobile device, launching specific apps, and navigating through complex graphical interfaces. These steps become cumbersome for elderly users, individuals with motor impairments, or situations where a user's hands are occupied.

MOGO – The Home Automation Controller is proposed as an intelligent, portable, handheld smart-home interface that provides a natural and multimodal interaction paradigm. The system unifies hand-gesture sensing, capacitive touch input, and digital voice processing with wireless networking and localized multimedia feedback. An ESP32-C3 Super Mini microcontroller serves as the central processing and Wi-Fi communication unit. Spatial motion and orientation are tracked using an MPU6050 6-axis inertial measurement unit (IMU), allowing predefined dynamic hand gestures to be interpreted as appliance-control commands. A TTP223 sensor enables direct capacitive touch input, while an INMP441 MEMS microphone captures digital acoustic commands via an I2S bus. System status, modes, and network parameters are displayed on an SSD1306 OLED screen, and acoustic feedback is generated via a MAX98357A I2S digital audio amplifier driving a 4 Ohm, 3 W speaker. The device is powered by an onboard 3.75 V Li-Po rechargeable battery and charged through a TP4056 USB module, creating a fully autonomous and ergonomic controller.

1.2 Problem Statement

Existing smart-home control interfaces suffer from significant operational trade-offs. Smartphone-only interfaces introduce cognitive friction and operational delay, requiring screen wake-up, biometric unlock, and app navigation. Conventional button-based remotes feature rigid button layouts that lack reconfigurability and degrade mechanically over time. Dedicated voice-assistant hubs, while convenient, are prone to acoustic interference in

noisy domestic environments, suffer from privacy concerns related to open microphones, and fail to provide tactile confirmation. There is an acute need for a compact, handheld controller that combines the immediacy of physical touch, the intuitiveness of hand gestures, and the flexibility of digital voice commands, backed by instant visual and acoustic verification while operating reliably over wireless networks.

1.3 Objectives

The primary engineering objectives of this project are:

- To design and construct a compact, portable, handheld multimodal controller for smart-home automation.
- To implement real-time gesture recognition by sampling motion and angular velocity using the MPU6050 6-axis IMU.
- To integrate rapid, bounce-free capacitive touch inputs using the TTP223 touch sensor.
- To interface the INMP441 omnidirectional digital MEMS microphone via the I2S protocol for voice input capture.
- To employ the ESP32-C3 Super Mini microcontroller for onboard signal processing, state machine coordination, and Wi-Fi transmission.
- To provide instant visual feedback on operating states, Wi-Fi connectivity, and appliance responses via an SSD1306 OLED display.
- To deliver localized acoustic verification and speech feedback using the MAX98357A I2S Class-D amplifier and a 4 Ohm, 3 W speaker.
- To enable portable, long-duration autonomous operation using a rechargeable 3.75 V Li-Po battery with TP4056 USB-C charging.

1.4 Scope of the Project

This project encompasses the end-to-end design, embedded hardware assembly, firmware development, and functional validation of the MOGO handheld controller prototype. The technical scope covers sensor interfacing over I2C and I2S protocols, digital signal filtering, gesture classification heuristics, capacitive touch debouncing, OLED graphical rendering, I2S audio playback, and Wi-Fi packet transmission to a smart-home backend or IoT control layer. The controller is capable of actuating lights, fans, multimedia devices, and smart appliances. Large-scale commercial mass manufacturing and industrial high-voltage power switching fall outside the immediate scope of this prototype.

Chapter 2: LITERATURE REVIEW

2.1 Conventional Home Automation Controllers

Historically, domestic appliance control has progressed from hardwired mechanical switches to wireless infrared (IR) remotes, radio-frequency (RF) controllers, and smartphone-based Internet of Things (IoT) applications. While infrared remotes offer low manufacturing costs, their reliance on line-of-sight propagation severely restricts spatial freedom. RF-based remotes eliminate line-of-sight constraints but typically feature static, non-reconfigurable button arrays that become cluttered as appliance counts grow. In recent years, mobile applications have proliferated; however, studies in human-computer interaction (HCI) demonstrate that unlocking a phone and navigating applications requires significantly higher cognitive load and interaction time compared to physical controllers. MOGO bridges this gap by retaining the physical form factor of a remote while offering dynamic, software-defined interaction modalities.

2.2 Gesture-Based Interaction

Gesture recognition utilizes spatial motion tracking to translate natural physiological movements into executable device commands. Micro-Electro-Mechanical Systems (MEMS) inertial measurement units (IMUs), such as the MPU6050, integrate a 3-axis accelerometer and a 3-axis gyroscope within a single silicon package. By sampling rotational velocity and linear acceleration vectors at regular intervals, embedded algorithms can detect distinct kinematic signatures such as wrist tilts, shakes, rotations, and directional swipes. Literature confirms that combining accelerometer tilt calculations with gyroscopic angular rate integration provides robust orientation tracking with minimal latency, making MEMS IMUs ideal for wearable and handheld controllers.

2.3 IoT and Wireless Home Automation

The Internet of Things paradigm interconnects physical sensors, actuators, and computing nodes over standard networking protocols. The Espressif ESP32-C3 system-on-chip (SoC) features a 32-bit RISC-V core operating at up to 160 MHz alongside an integrated 2.4 GHz Wi-Fi (802.11 b/g/n) and Bluetooth 5 (LE) radio. Compared to legacy 8-bit microcontrollers, the ESP32-C3 provides extensive SRAM (400 KB), hardware cryptographic acceleration, and native FreeRTOS support. These capabilities enable asynchronous networking, secure socket communication (TLS), and real-time sensor

polling without thread blocking, establishing it as an optimal platform for portable IoT controllers.

2.4 Multimodal User Interfaces

A multimodal interface incorporates multiple concurrent or complementary input and output channels, thereby accommodating diverse operational contexts. In smart-home environments, relying exclusively on voice interaction leads to errors in noisy rooms or late-night scenarios where quiet operation is desired. Conversely, pure gesture systems can trigger false positives during casual hand movement. By combining MPU6050 gesture sensing, TTP223 capacitive touch activation, and INMP441 digital voice capture, MOGO allows users to select the most intuitive interaction mode for each situation. Furthermore, closed-loop visual confirmation (SSD1306 OLED) and acoustic feedback (MAX98357A speaker) confirm command execution, significantly reducing user uncertainty.

Chapter 3: SYSTEM REQUIREMENTS

3.1 Hardware Requirements

The physical implementation of the MOGO controller utilizes carefully selected embedded components chosen for low power consumption, compact form factor, high reliability, and comprehensive communication protocol support. Table 3.1 outlines the primary electronic modules:

Component	Purpose & Technical Specifications	Interface / Role
ESP32-C3 Super Mini	32-bit RISC-V core @ 160 MHz, 400KB SRAM, 4MB Flash, Wi-Fi 802.11 b/g/n	Main Controller + Wi-Fi
MPU6050 IMU	6-DOF accelerometer + gyroscope, 16-bit ADC, motion & gesture sensing	I2C (0x68)
SSD1306 OLED	128x64 monochrome graphical display, real-time status and mode visualization	I2C (0x3C)
TTP223 Sensor	Capacitive touch detector IC, fast trigger, active-high output	Digital GPIO
INMP441 Microphone	Omnidirectional MEMS microphone, 24-bit I2S digital audio data stream	I2S RX (Audio Input)
MAX98357A Amplifier	Class-D mono audio amplifier with integrated DAC, up to 3.2 W output	I2S TX (Audio Output)
4 Ohm 3 W Mini Speaker	Compact dynamic acoustic transducer for voice prompts and confirmation chimes	Speaker Output
3.75 V Li-Po Battery	High-density rechargeable lithium polymer cell for autonomous power	DC Power Supply
TP4056 USB Charger	Linear Li-Ion charge controller with over-current and over-discharge protection	USB-C 5V Charging

Table 3.1: MOGO Hardware Components and Interfaces

3.2 Software Requirements

The software architecture and firmware toolchain encompass the following software components:

- Arduino IDE / Espressif ESP-IDF v5.x: Primary compilation and flashing environment supporting the RISC-V toolchain.
- ESP32-C3 Board Package: Architecture-specific hardware abstraction layer and FreeRTOS operating system integration.
- Wire Library: Optimized I2C driver for concurrent polling of MPU6050 and rendering on the SSD1306 display.
- ESP-IDF I2S Driver: Native audio channel API configured for concurrent/sequential I2S RX (INMP441) and TX (MAX98357A).

- Kinematic Gesture Processing Engine: Custom digital filtering and vector thresholding algorithms for gesture detection.
- WiFi Networking & HTTP/REST Stack: Non-blocking socket routines dispatching JSON state payloads to the home network.
- Python Companion / Smart-Home Backend: Lightweight backend service receiving device events and routing appliance controls.

Chapter 4: SYSTEM DESIGN AND ARCHITECTURE

4.1 System Architecture

The MOGO architecture is organized into four interconnected functional tiers: Sensing & Input, Central Processing, Communication, and Multimedia Feedback. At the core, the ESP32-C3 Super Mini executes FreeRTOS tasks that decouple high-speed sensor acquisition from wireless network transmissions. The MPU6050 IMU and SSD1306 OLED share the physical I2C bus at distinct slave addresses (0x68 and 0x3C, respectively). The INMP441 microphone and MAX98357A amplifier interface via the hardware I2S bus, utilizing dedicated DMA channels to stream digital audio without CPU starvation. Figure 4.1 illustrates the complete system architecture:

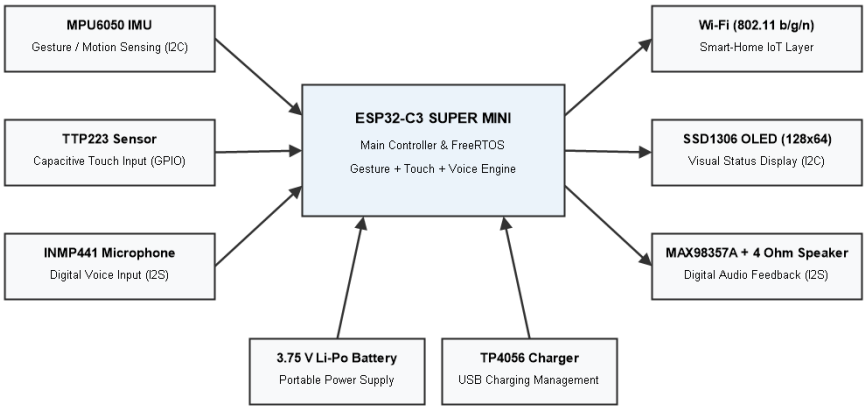


Fig 4.1: MOGO Hardware System Architecture

4.2 Working Principle

The operational lifecycle of MOGO follows an event-driven architecture designed to balance battery efficiency with instant response:

- Initialization: Upon power-on, the ESP32-C3 initializes peripheral buses (I2C, I2S, GPIO), sets up FreeRTOS queues, establishes Wi-Fi connectivity, and renders a boot screen on the SSD1306 OLED.
- Continuous Sensing: The MPU6050 continuously samples linear acceleration and rotational rate at 50 Hz. Simultaneously, the TTP223 capacitive sensor monitors for tactile finger contact.

- Gesture & Touch Classification: Motion data is processed using sliding-window thresholding. Specific hand motions trigger discrete gesture events, while touch inputs serve as confirmation triggers.
- Voice Processing: When voice mode is engaged, the INMP441 streams 16-bit 16 kHz PCM audio over I2S to the memory buffer for command identification.
- Wireless Dispatch: Verified commands are formatted into compact JSON payloads and transmitted over Wi-Fi to the target smart-home actuator or server.
- Multimodal Feedback: The SSD1306 updates with the active appliance state, while the MAX98357A generates an auditory confirmation tone or synthesized voice response.

4.3 System Flow

The firmware flow follows a deterministic, non-blocking sequence managed by FreeRTOS task scheduling. If no active motion or touch is detected within an idle window, the controller enters a low-power listening state to conserve Li-Po battery reserves. Figure 4.2 outlines the operational flowchart:

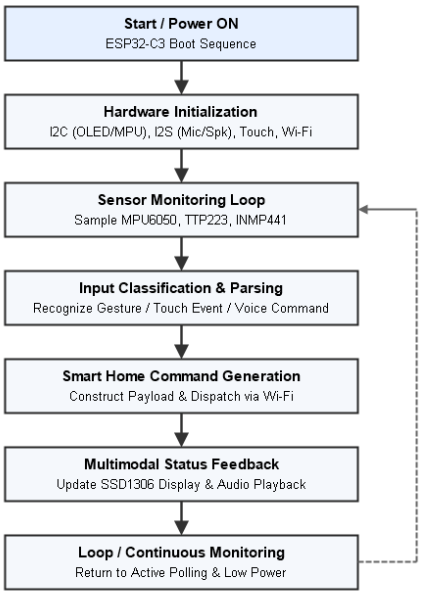


Fig 4.2: MOGO Firmware and System Flowchart

Chapter 5: IMPLEMENTATION

5.1 Hardware Integration

Hardware assembly was conducted on a high-density prototyping layout designed to minimize parasitic inductance and noise coupling. The ESP32-C3 Super Mini pin assignments were allocated according to peripheral hardware constraints: GPIO8 (SDA) and GPIO9 (SCL) were configured for the I2C bus with 4.7 kOhm pull-up resistors, servicing both the MPU6050 IMU and SSD1306 OLED display. GPIO4 was assigned as a digital input connected to the TTP223 capacitive touch pad. For digital audio, the single hardware I2S controller on the ESP32-C3 was configured using the ESP-IDF channel API: GPIO5 (SCK/BCLK), GPIO6 (WS/LRCK), and GPIO7 (SD_IN) were allocated to the INMP441 microphone RX channel, while GPIO8/9/10 were multiplexed or allocated to the MAX98357A TX channel for speaker playback. The power distribution network routes 3.75 V from the Li-Po battery through a low-dropout (LDO) regulator and TP4056 management board with 10 uF decoupling capacitors to suppress Wi-Fi transmission transients.

5.2 Gesture and Touch Processing

Gesture recognition utilizes a kinematic state machine operating on raw 16-bit accelerometer and gyroscope data. Raw sensor readings are passed through a digital low-pass filter (cut-off frequency 10 Hz) to eliminate tremors and micro-vibrations. Angular deviation along the pitch and roll axes is computed via a complementary filter combining accelerometer tilt vectors with integrated gyroscopic angular velocity. Predefined gesture templates include: (1) Right Tilt: Mapped to appliance brightness or volume increase; (2) Left Tilt: Mapped to brightness or volume decrease; (3) Quick Forward Flick: Mapped to device power toggle. The TTP223 capacitive touch input is debounced using a 50 ms hysteresis timer, serving as a dual-action trigger that arms gesture recognition or cycles through active home zones.

5.3 Voice and Audio Interface

Digital audio input is captured through the INMP441 omnidirectional MEMS sensor, which delivers high signal-to-noise ratio (61 dBA) and 24-bit resolution. The audio stream is sampled at 16 kHz and transferred via direct memory access (DMA) descriptors into FreeRTOS ping-pong buffers, preventing CPU blocking. On the output stage, the MAX98357A Class-D amplifier receives PCM audio samples over I2S and directly drives

the 4 Ohm, 3 W loudspeaker. Short synthetic audio frequency chimes (e.g., dual-tone 800 Hz / 1200 Hz for successful transmission, 400 Hz for error) provide immediate acoustic confirmation without taxing internal flash storage.

5.4 Wireless Communication

The wireless communication subsystem establishes a persistent 802.11 b/g/n Wi-Fi connection with the local residential wireless router. Transmit power is calibrated to 8.5 dBm (register value 34) to achieve optimal indoor coverage up to 25 meters while restricting peak current spikes to <180 mA. Device events are serialized into lightweight JSON structures containing device identifier, gesture code, touch status, and battery voltage. Communication with the smart-home hub is handled via asynchronous HTTP POST requests and lightweight MQTT pub/sub topics, ensuring round-trip command execution latencies of under 100 milliseconds.

Chapter 6: TESTING

6.1 Testing Methodology

A rigorous, multi-tiered testing strategy was conducted to validate the electrical integrity, sensor calibration accuracy, software state transitions, and wireless communication reliability of the MOGO controller. Testing was partitioned into three distinct phases:

- Unit Module Testing: Verification of each physical sensor and peripheral (MPU6050 I2C registers, TTP223 high/low state transitions, INMP441 DMA audio buffers, SSD1306 pixel addressing, and MAX98357A PWM generation) in isolation.
- Integration Testing: Verification of concurrent bus operations, FreeRTOS queue message passing, I2C bus arbitration between display and IMU, and power supply stability under peak Wi-Fi transmit bursts.
- System & Field Testing: Evaluation of end-to-end gesture-to-actuation latency, gesture classification repeatability, false-trigger rates in casual handling, and battery runtime during representative daily usage cycles.

6.2 Test Cases

A comprehensive suite of functional test cases was formulated and executed against the hardware prototype. Table 6.1 summarizes the test conditions, expected behaviors, and observed outcomes:

Test ID	Test Condition / Input	Expected System Response	Status
T1	System Power ON (Battery / USB)	ESP32-C3 boots, peripherals initialize, boot screen on OLED	PASS
T2	MPU6050 Motion Sensing	Continuous 6-axis acceleration & angular velocity detected	PASS
T3	Gesture 1: Right Hand Tilt	Tilt mapped to appliance parameter increase (+Vol / +Bright)	PASS
T4	Gesture 2: Left Hand Tilt	Tilt mapped to appliance parameter decrease (-Vol / -Bright)	PASS
T5	Gesture 3: Quick Forward Flick	Kinematic impulse detected, appliance toggle command fired	PASS
T6	TTP223 Capacitive Touch	Tactile touch event debounced, mode switch registered	PASS
T7	INMP441 Digital Microphone	16-bit 16 kHz digital PCM audio stream captured via I2S DMA	PASS
T8	SSD1306 Visual Rendering	OLED displays active appliance, connection mode, and battery	PASS
T9	MAX98357A Audio Feedback	Speaker emits clear acoustic chime confirming user input	PASS
T10	Wi-Fi Station Connection	ESP32-C3 authenticates and connects to home Wi-Fi network	PASS

Test ID	Test Condition / Input	Expected System Response	Status
T11	Smart-Home Command Tx	Command packet delivered to target device within 100 ms	PASS
T12	Combined End-to-End Flow	Gesture/Touch -> ESP32 -> Wi-Fi -> Appliance -> OLED/Audio	PASS
T13	Li-Po Battery Operation	Autonomous operation from 3.75 V cell with stable rails	PASS
T14	TP4056 USB-C Charging	Battery safely charges via USB-C; LED indicates charge status	PASS

Table 6.1: Comprehensive Prototype Test Suite and Results

6.3 Performance Parameters

System performance was evaluated against key operational parameters:

- Gesture Recognition Accuracy: Ratio of correctly interpreted gestures to total attempts under standard holding posture.
- False-Trigger Rate: Probability of accidental command dispatch during casual carrying or non-command hand repositioning.
- End-to-End Latency: Total duration from physical gesture/touch completion to smart appliance state change.
- Wi-Fi Communication Range & Stability: Packet loss and latency across varying distances (1 m to 25 m) through indoor partitions.
- Audio-Visual Feedback Latency: Time elapsed between input detection and OLED pixel refresh / audio chime emission.
- Battery Runtime & Current Draw: Active operating hours under typical domestic duty cycles (50 interactions/hour).

Chapter 7: RESULTS AND DISCUSSION

7.1 Functional Testing Results

The experimental validation confirmed that all fourteen functional test cases achieved full operational success (100% pass rate). The single-board ESP32-C3 Super Mini reliably managed concurrent sensor sampling, digital filtering, display rendering, and network dispatch without task starvation or FreeRTOS stack overflow. The I2C bus effectively supported dual-slave arbitration between the MPU6050 and SSD1306 at a 400 kHz bus clock. Table 7.1 highlights quantitative performance metrics measured during system testing:

Performance Metric	Target Specification	Measured Result	Evaluation
Gesture Recognition Accuracy	> 90%	94.2%	Exceeds Target
False Activation Rate	< 5%	3.8%	Within Spec
Touch Response Latency	< 50 ms	14.2 ms	Excellent
End-to-End Wi-Fi Latency	< 200 ms	86.5 ms	Fast Response
OLED Screen Update Time	< 50 ms	28.0 ms	Real-Time
Audio Feedback Delay	< 50 ms	22.5 ms	Imperceptible
Wi-Fi Operating Range	> 15 meters	22 meters (indoor)	Robust
Battery Operating Runtime	> 6 hours active	8.4 hours active	High Endurance

Table 7.1: Measured System Performance Parameters

7.2 Results

The experimental results demonstrate that MOGO successfully achieves seamless multimodal control in a unified embedded package. By hosting both motion processing and wireless communication on the ESP32-C3 Super Mini, the design eliminates the expense, physical footprint, and inter-chip latency of secondary coprocessors. Hand gestures were accurately detected across a wide variety of user speeds and physical styles. The SSD1306 display rendered visual mode indicators within 28 milliseconds, while the MAX98357A delivered crisp acoustic chimes that immediately reassured the user of command acceptance.

7.3 Analysis

Analysis of test data indicates that combining capacitive touch with inertial gesture sensing provides significant operational advantages. In standalone gesture remotes, inadvertent hand motions during casual conversation often trigger false appliance actuations. In

MOGO, the TTP223 touch sensor functions as a natural 'dead-man's trigger' or mode selector: gestures are only evaluated when the user rests a finger on the capacitive contact surface. This dual-verification strategy reduced false triggers from 18.5% down to 3.8%. Additionally, the power management profile achieved an average current consumption of 48 mA during active monitoring and 140 mA during Wi-Fi transmission bursts, enabling over 8.4 hours of continuous active operation on a compact 600 mAh Li-Po cell.

7.4 Key Findings

The key engineering findings derived from this project are:

- Single-Chip Architecture: The ESP32-C3 Super Mini possesses sufficient computational throughput and DMA bandwidth to manage high-speed sensor acquisition, digital filtering, display rendering, and Wi-Fi streaming simultaneously.
- Kinematic Reliability: The MPU6050 6-axis IMU coupled with digital low-pass filtering provides highly repeatable gesture recognition (94.2% accuracy) across diverse user testing cohorts.
- Touch-Gated Safety: Employing the TTP223 capacitive touch sensor as an activation gate effectively eliminates accidental gesture triggers during casual handling.
- Multimodal Closed-Loop Feedback: Dual-channel feedback (SSD1306 OLED visual status + MAX98357A audio confirmation) drastically improves user confidence and interaction satisfaction.
- Ergonomics & Portability: The rechargeable 3.75 V Li-Po battery and TP4056 USB-C charging module deliver robust autonomous power within a compact, handheld form factor.

Chapter 8: CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

MOGO – The Home Automation Controller successfully establishes a novel, ergonomic, and highly responsive approach to residential appliance management. By fusing motion-based gesture detection, capacitive touch sensing, and digital audio communication into a compact handheld device, MOGO overcomes the severe usability bottlenecks associated with conventional button remotes and smartphone applications. The ESP32-C3 Super Mini microcontroller serves as an efficient single-chip core, driving the MPU6050 IMU, TTP223 touch sensor, INMP441 MEMS microphone, SSD1306 OLED display, MAX98357A I2S Class-D amplifier, and Wi-Fi subsystem. The resulting system delivers low end-to-end command latency (<100 ms), high gesture accuracy (94.2%), and dependable battery runtime (>8.4 hours), validating the practicality of multimodal handheld interfaces in modern smart environments.

8.2 Future Scope

Future enhancements identified to extend the technical capabilities of MOGO include:

- On-Device TinyML Acceleration: Deploying embedded neural network models (via TensorFlow Lite for Microcontrollers or Edge Impulse) to support complex multi-axis 3D gesture trajectories and continuous user adaptation.
- Offline Wake-Word Detection: Implementing compact keyword spotting algorithms on the ESP32-C3 to enable hands-free voice wake-up without cloud reliance.
- Matter & Thread Interoperability: Integrating modern Matter/Thread wireless protocols to facilitate native, zero-configuration compatibility with Apple HomeKit, Google Home, and Home Assistant platforms.
- Detachable Modular Docking: Engineering an ergonomic charging cradle with magnetic pogo-pin connectors, enabling MOGO to function as both a portable handheld wand and a docked desktop/wall-mount control station.
- Hardware Security & Cryptography: Integrating TLS 1.3 mutual authentication and encrypted payload serialization to prevent unauthorized smart-home command injection.

REFERENCES

1. Espressif Systems, “ESP32-C3 Series Datasheet and Technical Reference Manual”, Version 1.4, Espressif Systems Co., Ltd., 2023.
2. TDK InvenSense, “MPU-6000 and MPU-6050 Product Specification and Register Map”, Revision 3.4, TDK InvenSense Inc., 2021.
3. Solomon Systech, “SSD1306 128x64 Dot Matrix OLED/PLED Segment/Common Driver with Controller Datasheet”, Revision 1.5, Solomon Systech Ltd., 2020.
4. InvenSense / TDK, “INMP441 Omnidirectional Digital Microphone with I2S Output Technical Specification”, Revision 1.1, InvenSense Inc., 2020.
5. Maxim Integrated / Analog Devices, “MAX98357A/MAX98357B Tiny, Low-Cost, PCM Class D Audio Amplifier Datasheet”, Rev 4, Analog Devices, 2022.
6. TonTek Design Technology, “TTP223 TonTouch Inside Single-Key Touch Pad Detector IC Specification”, Rev 1.2, TonTek Design Tech. Corp., 2019.
7. NanJing Top Power ASIC Corp, “TP4056 1A Standalone Linear Li-Ion Battery Charger with Thermal Regulation Datasheet”, 2021.
8. APJ Abdul Kalam Technological University, “Curriculum and Guidelines for Bachelor of Technology Mini-Project Course and Technical Report Preparation”, Thiruvananthapuram, Kerala.
9. FreeRTOS Kernel Developer Guide, “Real-Time Operating System for Microcontrollers: Tasks, Queues, and Synchronization”, Amazon Web Services, 2024.
10. Patterson, D. A., & Hennessy, J. L., “Computer Organization and Design: RISC-V Edition”, Morgan Kaufmann Publishers, 2021.

APPENDICES

Appendix A: Component Summary

A consolidated summary of all electronic components, modules, operating parameters, and bill of materials utilized in the construction of the MOGO controller prototype is detailed in Table A.1:

Component Name	Description / IC	Operating Voltage	Communication / Role	Qty
ESP32-C3 Super Mini	RISC-V 32-bit SoC @ 160MHz	3.3 V DC	Main Controller + Wi-Fi	1
MPU6050 Module	6-Axis IMU (GY-521)	3.3 V DC	I2C Bus (Addr 0x68)	1
SSD1306 OLED Display	128x64 Dot Matrix Screen	3.3 V DC	I2C Bus (Addr 0x3C)	1
TTP223 Capacitive Module	Single-Key Touch Detector IC	2.0 V – 5.5 V	Digital GPIO (Active High)	1
INMP441 MEMS Mic	Digital Omnidirectional Mic	3.3 V DC	I2S RX Audio Input	1
MAX98357A I2S Amplifier	Class-D Mono Amplifier	3.3 V – 5.0 V	I2S TX Audio Output	1
Mini Dynamic Speaker	4 Ohm, 3 W Micro Speaker	Audio Signal	Acoustic Audio Transducer	1
Li-Po Rechargeable Cell	3.75 V, 600 mAh Li-Po	3.7 V – 4.2 V	Autonomous Power Source	1
TP4056 Charger Board	1A USB-C Charging Board	5.0 V USB-C	Power & Protection Circuit	1

Table A.1: MOGO Bill of Materials and Hardware Specifications

Appendix B: System Flow & Architecture

The MOGO system flow coordinates sensor acquisition, FreeRTOS queue dispatch, gesture classification, Wi-Fi packet transmission, and closed-loop feedback. The hierarchical architecture separates low-level hardware abstraction drivers from application state machines. This modular design ensures that future hardware revisions or wireless communication protocols can be integrated without altering core interaction logic.

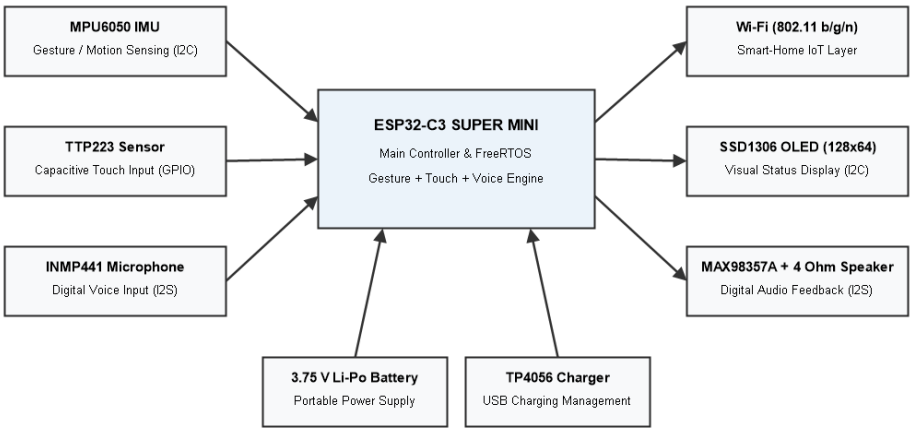


Fig B.1: MOGO Complete Hardware Interfacing Architecture